The Universal Warmup Path: Many Routes, One Compass

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Abstract

Sampler transitions are local, but their efficiency is governed by global posterior geometry. Warmup bridges the two by tuning the step size, metric, and related controls. Although these adaptation mechanisms are individually well studied, their orchestration is still commonly encoded through fixed schedules and heuristic choices: users preselect the metric family, compute budget, and response to incompatible geometry. We recast that orchestration as evidence-based routing within a declared budget. The Universal Warmup Path uses one sampler-independent compass, $\Sigma_\pi = \mathrm{Cov}_\pi(X)$, while allowing each route to deploy its own metric. Its discipline—gather evidence, act, wait, or refuse—is sampler-independent; we evaluate one gradient-based HMC-family realization using scalar gates. The confidence-aware design gives route-attractor guarantees, while a finite-transcript bound limits what confined evidence can establish. When local and global evidence agree, the path deploys an efficient local metric; disagreement fails loudly through reparameterization advice or population handoff, never certifying global coverage. Across the cells we tested, automatic warmup outperformed the predeclared Fisher low-rank primary, with geometric-mean ESS-per-gradient ratios of 1.409–2.451 on the ill-conditioned suite and 1.131–1.951 on German credit. It also improved ESS per gradient over the predeclared fixed-schedule diagonal control while meeting the same declared post-sampling quality criterion. The proposed path points toward a coherent theory of sampler hyperparameter tuning, connecting local adaptation, global geometry, and explicit refusal in practical controllers.

1 Introduction

Hamiltonian Monte Carlo (HMC) uses local evaluations of a target to explore a global distribution [Duane et al., 1987, Neal, 2011]. Its metric and step size determine whether those local moves resolve the typical set efficiently; NUTS adapts trajectory length around the same geometry [Hoffman and Gelman, 2014]. Warmup must learn that geometry from the least trustworthy part of the run: correlated trajectories before representativeness has been established.

This paper treats warmup as a hybrid route-plus-metric dynamical system. Each estimator branch a has a population functional

$$G_a^\star = T_a(\pi).$$

Examples include diagonal scale, a pooled-within low-rank correction, and a between-means low-rank correction. The two low-rank branches may share a reported label while targeting different population objects. The proposed confidence-aware latch selects a branch only after its confidence set clears that branch's eligibility region; conditional on a stable route episode, the population metric map may then attract the iterate to $G^\star_a$.

Four questions remain distinct throughout. *Attraction* asks whether a fixed route contracts. *Structural identification* asks whether the transcript resolves a rank or subspace. *Representativeness* asks whether the transcript has missed important regions. *Utility* asks whether an identified structure improves compute-adjusted sampling. None implies the next.

The resulting warmup path is universal in a procedural sense: gather evidence, promote only when the evidence supports a route, and refuse unsupported geometry. Under a declared constant-metric adequacy criterion, funnels and scale-coupling targets route to reparameterization. Genuinely regional mixtures route instead to the companion population/ensemble method over regional attractors. Refusal is therefore an intended outcome, not a failed metric fit.

Our contributions are a route-indexed attractor theorem with explicit finite- window error terms; a Markov-transcript uncertainty construction and its operator consequences; a finite-horizon local-transcript information bound; and a scalar-gate implementation that keeps identification, consistency, and utility separate. The controlled Gaussian-mixture study holds the marginal spectrum fixed while the recorded within/between evidence and route change, exposing both the value and the limits of the gate. The empirical corpus separates a predeclared Fisher low-rank primary, a diagonal control, and a historical shared-step implementation comparison.

Section 2 fixes the metric convention, and Section 3 gives the practical route tree. Sections 4–6 develop the local guarantees and limits. Section 7 reports the empirical comparisons; implementation, related work, and scope follow, with proofs and supporting studies in the appendices.

2 Metric convention and warmup routes

Let $\pi(x) \propto \exp\{\mathcal{L}(x)\}$ on $\mathbb{R}^d$. We use the *inverse-mass* convention

$$p \sim \mathcal{N}(0, G^{-1}), \qquad K(p) = \tfrac{1}{2} p^\top G p. \tag{1}$$

Thus BlackJAX's `inverse_mass_matrix` is G. For $x \sim \mathcal{N}(\mu, \Sigma)$, Hamilton's equations give $\ddot{x} = -G\Sigma^{-1}(x-\mu)$, so the covariance-matching choice is $G^\star = \Sigma$.

For a general target with finite second moments, define the universal sampler-independent covariance reference

$$\Sigma_\pi = \mathrm{Cov}_\pi(X).$$

It is a common reference, not a common route estimand: a Fisher, masked, regularized, or finite-rank route need not target Σ_π.

For a route a, let $G^\star_a = T_a(\pi)$ and define the affine-invariant residual

$$R_a(G) = G^{-1/2} G^\star_a G^{-1/2}.$$

The tangent from G toward $G^\star_a$ is represented by $\log R_a(G)$ in the whitened frame. This is not interchangeable with the log-Euclidean difference $\log G^\star_a - \log G$ unless the matrices commute. For the recursion below we work in a bounded SPD chart $\theta = \log G$; the route-specific moment-to-metric map is part of T_a, not silently identified with either matrix logarithm.

Classical window adaptation updates step size in fast and slow intervals, while learning covariance only in expanding, memoryless slow intervals [Carpenter et al., 2017, Stan Development Team, 2026]. Fisher-divergence preconditioning adds a low-rank-plus-diagonal option [Seyboldt et al., 2026]. Automatic warmup must decide not only how to estimate within a branch, but when the transcript makes that branch eligible.

3 Controller and evaluation protocol

The evaluated implementation starts diagonal and constructs its decision schedule from dimension d, chain count M, and total gradient budget $\mathcal{B}$. For NUTS it assigns $\lfloor \mathcal{B}/(20M) \rfloor$ transitions to each chain, using 20 as a conservative gradients-per-transition allowance; a fixed-L kernel uses its declared integration length in place of 20. The dimension-derived rank capacity and pooled support floor are

$$k_{\mathrm{cap}} = \min\{50, \max(\lfloor d/2 \rfloor, 1)\}, \qquad N_{\min} = 8(k_{\mathrm{cap}} + 1).$$

After a harmless one-step initialization window, the first evidence-bearing window has $n_1 = \lceil N_{\min}/M \rceil$ transitions per chain. Metric windows then have nominal $1.5\times$ growth. When another grown window would not fit before the step-size-only phase, the last slow window instead absorbs the remaining metric transitions. The final 15% adapts step size only. These boundaries are fixed before sampling begins.

At each prescribed metric-window endpoint, the controller finalizes the current M-chain buffer, recomputes the evidence, and resets the buffer. The route-specific conjunction combines the W or T structural tests, the R^2 metric-fixability check, and enough remaining budget for the selected rank and step-size readaptation. A supported W route deploys a low-rank metric from pooled, per-chain-centered within-chain information. A supported T route deploys a between-means low-rank correction. Unsupported or inconclusive evidence leaves the metric diagonal and advances to the next larger scheduled window. Decisions occur only at endpoints: the evaluated implementation neither terminates within a window nor moves a boundary in response to the data.

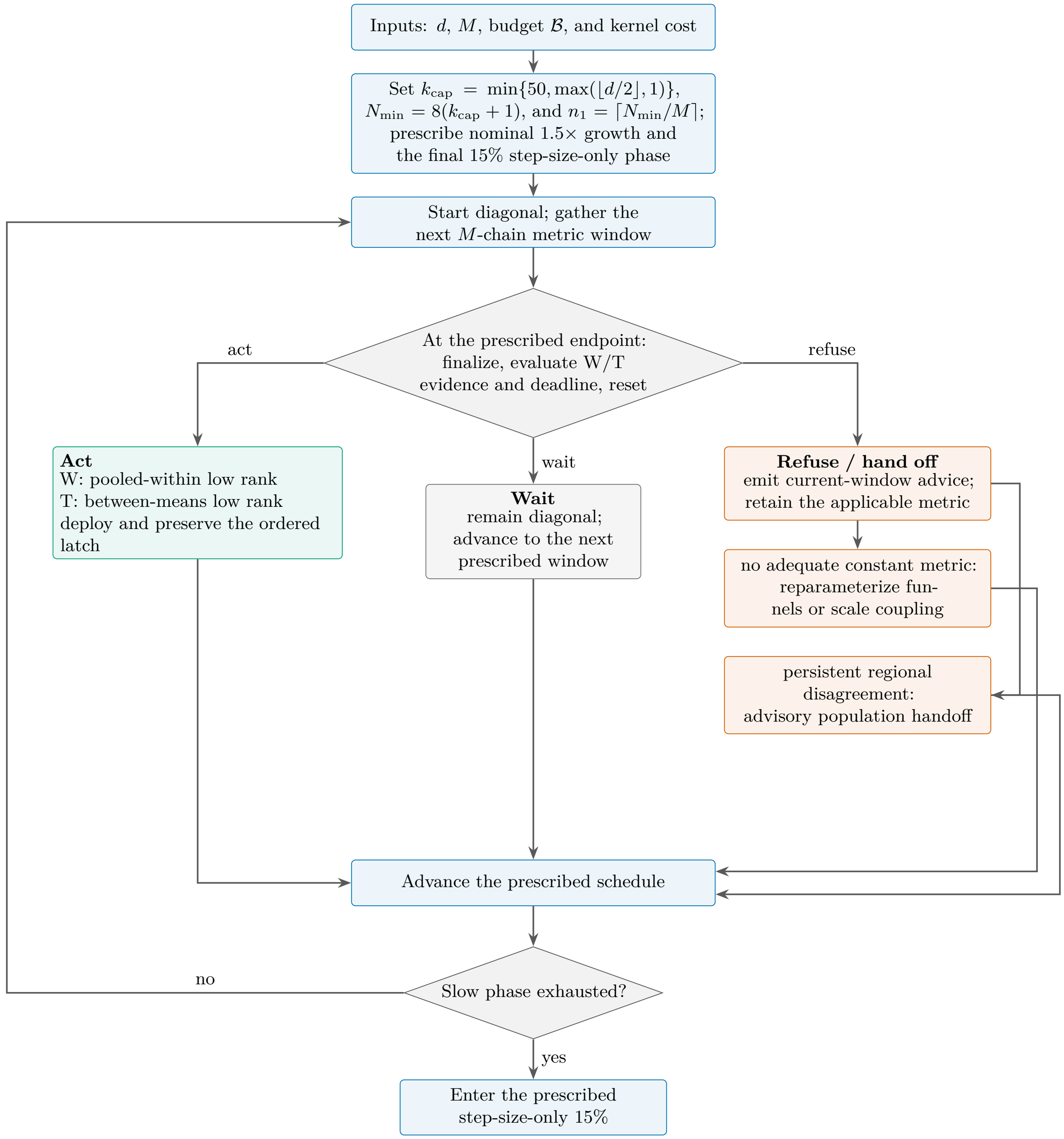

Figure 1: The evaluated warmup path. Setup fixes every window boundary before execution, and evidence is evaluated only at metric-window endpoints. Every outcome continues the prescribed scan: acting preserves the selected route, waiting advances while diagonal to the next prescribed window, and the two advisory leaves retain the applicable metric. Only exhaustion of the metric-adapting phase enters the final step-size-only 15%. This is not an early-stopping or change-point rule.

Persistent regional disagreement can coexist with a W-selected, within-chain-conditioned metric. It produces a current-window advisory handoff to the companion population/regional ensemble; it never establishes global exploration, which is reported as `not_established`. Reparameterization is a separate no-adequate-constant-metric route. Both advisory outputs are recomputed at later endpoints and may clear while the scan continues. A companion local-trajectory sampler may consume the selected local metric, while the population method owns discovery, exchange, and weighting across regional attractors. The theorem-level design below promotes only after confidence-ball inclusion; the evaluated implementation realizes the same act–wait–refuse discipline with scalar gates, not with those confidence objects.

Estimator, window schedule, metric-buffer policy, and step-size controller are independent engineering choices. The Fisher-HMC reference prescription in this paper is specifically the 30/55/15 schedule of Seyboldt et al. [2026]: it uses $L = 10$ and then $L = 80$ refresh/memory periods, periodically wipes older draws, and reinitializes step size once at the start of phase 2. It is not a generic description of every `nutpie` version. Stan's expanding slow windows are successively doubled in the version 2.39 CmdStan User's Guide [Stan Development Team, 2026].

The empirical schedule evidence compares configuration bundles and does not isolate the causal effect of window shape; its limited reseed ablation does not establish Stan–nutpie equivalence. Figure 5 in Appendix C records the schedule prescriptions and one executed trace.

4 Route-indexed attractor dynamics

Let η_k be the target-population routing object at the current metric, including the population moments and whitening transformation declared by the candidate route. Let C_k be its confidence set at the end of window k and H_b the eligibility region for promotion to route b. Coverage of C_k must pay for starting-law, within-window adaptation, whitening, sampling, and regularization errors. Routes form an ordered, one-transition latch: once promoted, the controller does not demote during that route episode.

Lemma 1 (Simultaneous soundness of the confidence-aware latch)**.** *Let $\mathcal{K}$ be the fixed, declared set of potential decision windows. Suppose*

$$\Pr(\eta_k \notin C_k \mid \mathcal{F}_{k-1}) \le \delta_k$$

for every $k \in \mathcal{K}$, whether or not a promotion is ultimately attempted. Promote to b only when $C_k \subset H_b$. Then, with probability at least $1 - \sum_{k\in\mathcal{K}} \delta_k$, every promotion made by the latch is supported by the true routing estimand at the time of promotion.

The lemma certifies only the declared target-population estimand at the promotion time. Persistent optimality and global target coverage require separate arguments. A fixed-window Wald set is not automatically a valid repeated-use sequence; a deployed sequential latch needs confidence spending or an anytime-valid construction.

After promotion, consider a route episode on branch a. Let $s(\theta)$ be its population routing statistic, g the router, $h_a(\theta)$ the population update field, and $\theta_a^\star = \log G_a^\star$. Write

$$F_a(\theta) = \theta + \alpha h_a(\theta), \qquad \theta_{k+1} = \widehat{F}_{k,a}(\theta_k). \tag{2}$$

Theorem 2 (Local robust attractor of a route episode)**.** *Let Θ be the bounded log-SPD chart used by the controller. Assume $R_0 > 0$, $\theta_a^\star \in \Theta$, and $\overline{B}_F(\theta_a^\star, R_0) \subset \Theta$, where $\overline{B}_F$ denotes* *a closed Frobenius-norm ball. Assume also that* $s(\theta_a^\star) \in \operatorname{int}(\mathcal{S}_a)$*, where* $\mathcal{S}_a = \{s : g(s) = a\}$*. Suppose* $h_a(\theta_a^\star) = 0$ *and, on* $\overline{B}_F(\theta_a^\star, R_0)$*, for constants* $\mu > 0$ *and* $L_h > 0$*,*

$$\langle \theta - \theta_a^\star, h_a(\theta) \rangle_F \le -\mu \|\theta - \theta_a^\star\|_F^2, \qquad \|h_a(\theta)\|_F \le L_h \|\theta - \theta_a^\star\|_F.$$

Let $0 < \alpha < 2\mu/L_h^2$ *and* $q = (1 - 2\alpha\mu + \alpha^2 L_h^2)^{1/2} < 1$*. For every* $k < K$*, suppose candidate statistics and updates are constructed before routing, and define the joint error event*

$$\begin{aligned}\mathcal{E}_k = \{&\|\widehat{s}_k - s(\theta_k)\|_{\mathcal{S}} > r_k^s\} \\ &\cup \{\|\widehat{F}_{k,a}(\theta_k) - F_a(\theta_k)\|_F > e_k\}, \qquad \Pr(\mathcal{E}_k \mid \mathcal{F}_{k-1}) \le \delta_k.\end{aligned}$$

Here $\|\cdot\|_{\mathcal{S}}$ *is the declared statistic-space norm. Let* s *be* L_s*-Lipschitz from the ball to that norm, let* $\kappa = \operatorname{dist}\{s(\theta_a^\star), \partial\mathcal{S}_a\} > 0$*, with distance in the same norm. Subject to the ordered latch, the observable rule selects or promotes candidate branch* a *whenever the following inclusion holds, and otherwise abstains:*

$$\overline{B}_{\mathcal{S}}(\widehat{s}_k, r_k^s) \subset \mathcal{S}_a.$$

For every $k < K$*, assume* $L_s R_0 + 2r_k^s < \kappa$ *and* $e_k \le (1-q)R_0$*; the derived margin is sufficient for the observable inclusion on the good event. If* $\|\theta_0 - \theta_a^\star\|_F \le R_0$ *and the latch is unset or already correctly set to* a*, then on the simultaneous good event the router selects* a*, the iterates remain in the ball, and*

$$\|\theta_K - \theta_a^\star\|_F \le q^K \|\theta_0 - \theta_a^\star\|_F + \sum_{j=0}^{K-1} q^{K-1-j} e_j. \tag{3}$$

This event has probability at least $1 - \sum_{k<K} \delta_k$*.*

The finite-window update-error budget is deliberately explicit:

$$e_k \le e_k^{\text{start}} + e_k^{\text{step}} + e_k^{\text{white}} + e_k^{\text{sample}} + e_k^{\text{reg}}. \tag{4}$$

These terms cover the starting law, adaptive step size, estimated whitening, sampling fluctuation, and regularization. Controlled-Markov stability, convergence, and ergodicity results provide asymptotic ingredients under conditions such as drift, containment, and diminishing adaptation; they do not themselves provide route-specific finite-window high-probability bounds for e_k^{start} or e_k^{step}. Invariant kernels alone do not remove transient or state-dependent adaptation bias [Andrieu et al., 2005, Fort et al., 2016, Roberts and Rosenthal, 2007]. The scalar-gate implementation has not calibrated or verified the five component bounds in (4). The theorem applies once those route-specific bounds have been established.

5 Uncertainty and safe structural gates

5.1 Universal covariance and route identification

For M equal-length chains with n recorded states, write $\bar{x}_m$ for chain m's mean and $\bar{x}$ for the grand mean, and define

$$W = \frac{1}{M(n-1)} \sum_{m=1}^{M} \sum_{t=1}^{n} (x_{mt} - \bar{x}_m)(x_{mt} - \bar{x}_m)^\top,$$

$$B = \frac{n}{M-1}\sum_{m=1}^{M}(\bar{x}_m - \bar{x})(\bar{x}_m - \bar{x})^\top.$$

If S is the ordinary covariance of all Mn states about $\bar{x}$, then

$$(Mn-1)S = M(n-1)W + (M-1)B. \tag{5}$$

This finite-transcript identity is exact: it requires neither independence nor stationarity. It remains exact after one common realized linear projection A, with S, W, B replaced by $ASA^\top, AWA^\top, ABA^\top$. A projection selected from the same data is still algebraically descriptive, but calibrated post-selection inference requires cross-fitting, selective inference, or a simultaneous operator bound. Separately whitening S, W, and B does not preserve (5).

Now hold M fixed and let $n \to \infty$. Suppose each chain's empirical first and second moments converge to those of a law ν_m. Then S converges to the covariance of the equal aggregate mixture

$$\bar{\nu} = \frac{1}{M}\sum_{m=1}^{M}\nu_m.$$

This limit identifies Σ_π only when $\bar{\nu}$ is second-moment representative of π. Agreement among chains is not enough: a clean ensemble confined to one basin can converge to the wrong regional covariance.

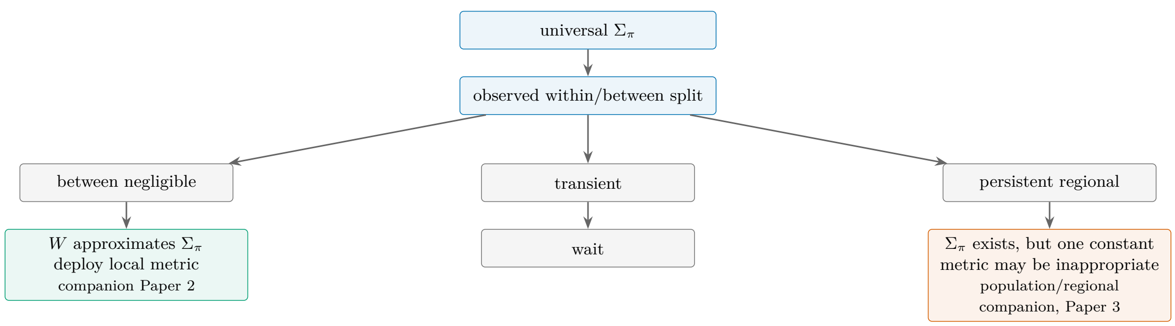

Figure 2: Practical interpretation of the universal covariance reference through the observed within/between split. The leaves scope metric use and the next method; they do not certify discovery or coverage of unseen regions.

Proposition 3 (When covariance identifies a route)**.** *Let $\mathcal{P}$ be a declared target class with finite covariance. A route functional T_a is identified by covariance on $\mathcal{P}$ if and only if it is constant on covariance fibres:*

$$\mathrm{Cov}_{\pi_0}(X) = \mathrm{Cov}_{\pi_1}(X) \quad\Longrightarrow\quad T_a(\pi_0) = T_a(\pi_1) \qquad (\pi_0, \pi_1 \in \mathcal{P}).$$

Equivalently, there is a map F_a on the covariance range of $\mathcal{P}$ such that $T_a(\pi) = F_a(\Sigma_\pi)$. This condition preserves the route-indexed target $G_a^\star = T_a(\pi)$ rather than replacing it by a universal metric.

The Fisher route gives a useful full-rank special case. Let $s(x) = \nabla \log \pi(x)$ and $J_\pi = E_\pi\{s(X)s(X)^\top\}$.

Proposition 4 (Full-rank Fisher covariance bound)**.** *Suppose π has a smooth positive density, finite SPD covariance Σ_π, square-integrable score, and Stein boundary conditions. Then*

$$E_\pi\{(X-\mu)s(X)^\top\} = -I, \qquad J_\pi \succeq \Sigma_\pi^{-1}.$$

For the unregularized, no-cutoff, full-rank affine-invariant Fisher population functional, with # denoting the SPD geometric mean,

$$G_F = \Sigma_\pi \# J_\pi^{-1}, \qquad \Sigma_\pi^{-1/2} G_F \Sigma_\pi^{-1/2} = (\Sigma_\pi^{1/2} J_\pi \Sigma_\pi^{1/2})^{-1/2} \preceq I.$$

Equality holds precisely in the affine-score, hence Gaussian, case under these assumptions [Dembo et al., 1991].

This proposition does not identify the deployed finite-rank, masked, regularized Fisher route from covariance. For example, a standard Gaussian and a variance-one Student-t_ν law have the same covariance, whereas their one-dimensional location Fisher informations are respectively 1 and $\nu(\nu+1)/\{(\nu-2)(\nu+3)\} > 1$ for $\nu > 2$. Their full-rank Fisher functionals therefore differ.

5.2 A route-specific Markov central limit theorem

For route a, collect the quadratic and score observables actually used by that branch in a vector $f_a(X_t)$, with mean η_a. For a stationary, geometrically ergodic frozen-kernel chain with sufficient moments, the multivariate Markov-chain CLT gives

$$\sqrt{n}(\widehat{\eta}_a - \eta_a) \Rightarrow \mathcal{N}(0, \Omega_a), \qquad \Omega_a = \sum_{h=-\infty}^{\infty} \mathrm{Cov}_\pi\{f_a(X_0), f_a(X_h)\}. \tag{6}$$

The full long-run covariance (LRV) depends on the functionals in f_a; a single scalar ESS cannot be substituted into an iid Wishart law [Jones, 2004, Vats et al., 2018]. If ϕ_a maps moments to a symmetric residual, let J_a be the derivative of $\mathrm{vech} \circ \phi_a$ and let

$$u = \mathrm{vech}(\widehat{R}_a - R_a),$$

where vech retains one copy of each off-diagonal. The delta method gives root-n limit covariance $\Gamma_a = J_a \Omega_a J_a^\top$ for $\sqrt{n}\,u$. Let Q be the oracle population support: its orthonormal columns span $\mathrm{range}(\Gamma_a)$. Set $z = Q^\top u$ and require $p = \mathrm{rank}(\Gamma_a) \geq 1$. If Γ_a is singular, assume $(I - QQ^\top)u = o_p(n^{-1/2})$, or that the residual lies exactly on the declared support. Require $Q^\top \widehat{\Gamma}_a Q$ to be positive definite and consistent on this support. For one fixed window, an illustrative Wald construction is

$$r_{\alpha,n} = \left\{ \frac{2\chi^2_{p,1-\alpha} \lambda_{\max}(Q^\top \widehat{\Gamma}_a Q)}{n} \right\}^{1/2}. \tag{7}$$

Indeed, for symmetric A, $\|A\|_F \leq \sqrt{2}\|\mathrm{vech}(A)\|_2$, which accounts for the factor 2 inside the squared radius and yields a Frobenius, hence operator, radius asymptotically. The construction requires a consistent multivariate LRV estimator and regularity for the route map. Estimating Q or its rank from the same window requires a separate justification. With M independent frozen chains, their contributions may scale as $1/M$; chains coupled through shared adaptation are not automatically independent. Non-asymptotic matrix concentration is another route only under explicit boundedness, tail, and dependence assumptions [Neeman et al., 2024].

Theorem 5 (Consequences of an operator-confidence event)**.** *Let R_a and $\widehat{R}_a$ be symmetric population and estimated residuals, with eigenvalues in descending order, and suppose $\|\widehat{R}_a - R_a\|_{\mathrm{op}} \le r$.*

1. *Weyl's inequality gives $|\widehat{\lambda}_j - \lambda_j| \le r$ for every j.*

2. *For a declared population cutoff τ and $1 \le k < d$, if $\widehat{\lambda}_k - r > \tau$ and $\widehat{\lambda}_{k+1} + r < \tau$, then exactly k population eigenvalues exceed τ. Otherwise this interval gate abstains.*

3. *Let V and $\widehat{V}$ span the matched population and estimated top-k eigenspaces (or a declared isolated eigenvalue cluster), and let Δ be the minimum population separation between that cluster and its complement. If $\Delta > 2r$, then $\|\sin\Theta(\widehat{V}, V)\|_{\mathrm{op}} \le 2r/\Delta$.*

4. *If $\lambda_{\min}(R_a) \ge m > r$, then*

$$\|\log \widehat{R}_a - \log R_a\|_{\mathrm{op}} \le \log\left(\frac{m}{m-r}\right) \le \frac{r}{m-r}.$$

A metric-error claim additionally requires a route-specific perturbation bound for the moment-to-metric map T_a.

These are standard operator, subspace, and SPD-log consequences [Bhatia, 1997, Yu et al., 2015, Higham, 2008]. They show exactly which margins a confidence-aware gate needs. The deployed cosine statistic Ψ is a deterministic agreement check between replicated directions: high agreement is compatible with shared whitening bias and does not establish accuracy or coverage.

5.3 The restricted role of BBP calibration

For iid Gaussian observations from the spiked covariance model with aspect ratio $\gamma = d/n$, a population spike separates when $\ell > 1 + \sqrt{\gamma}$, while the null sample-noise edge is $(1 + \sqrt{\gamma})^2$ [Baik et al., 2005, Baik and Silverstein, 2006, Paul, 2007]. This asymptotic PCA calculation is a useful scale calibration. It is not an exact edge for Markov draws, a finite-sample if-and-only-if statement, or an unrestricted information limit. In particular, subthreshold alternatives can retain nontrivial testing power [Onatski et al., 2013].

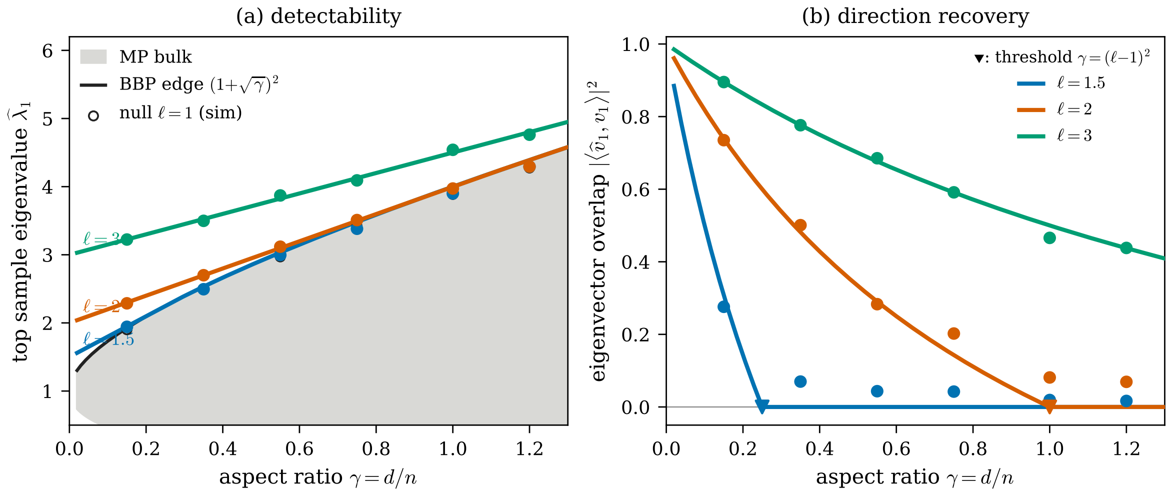

Figure 3: Iid Gaussian spiked-PCA calibration ($d = 200$). The curves show the asymptotic sample eigenvalue and eigenvector overlap; markers are fixed-seed simulations. The population threshold is $1 + \sqrt{\gamma}$, and the null sample edge is $(1 + \sqrt{\gamma})^2$. The figure is not a Markov-chain gate.

5.4 Controlled GMM at fixed marginal geometry

The controlled target is a five-dimensional, two-component Gaussian mixture with weights $\pi_1 = 0.30$, $\pi_2 = 0.70$, $v = (1, 1, 0, 0, 0)^\top/\sqrt{2}$, and $a = 21$. Its means and shared component covariance are

$$\mu_1 = -\pi_2\delta v, \qquad \mu_2 = \pi_1\delta v, \qquad \Sigma_w = I_5 + \beta vv^\top,$$

where

$$\delta^2(\mathrm{SR}) = \frac{\mathrm{SR}^2(1+a)}{1+\pi_1\pi_2\mathrm{SR}^2}, \qquad \beta = a - \pi_1\pi_2\delta^2.$$

Thus $\Sigma_{\text{marginal}} = \Sigma_w + \pi_1\pi_2(\mu_2 - \mu_1)(\mu_2 - \mu_1)^\top = I_5 + avv^\top$, so the marginal diagonal-whitened top eigenvalue is 1.913 throughout the separation sweep. Separation is indexed by $\mathrm{SR} = \delta/s(\delta)$, where δ is the distance between component means and $s(\delta) = \{v^\top\Sigma_w v\}^{1/2}$ is the within-component standard deviation along the separation direction v. Equation (5) therefore gives an exact finite-transcript partition for this target. With latent component labels, the usual within-component plus label-mean-scatter identity gives the corresponding population partition.

The current sweep keeps the marginal spike fixed at 1.913 while the recorded within/between evidence changes. Low-rank routing increasingly gives way to diagonal routing and an advisory population handoff. The route boundary is therefore not explained by a changing marginal spectrum, but the recorded summaries do not isolate a unique occupation mechanism. The handoff is recomputed from the current window and is non-monotone; it is neither a latch nor evidence of persistent global exploration. In every row, `global_exploration` remains `not_established`.

Compute changes when the disagreement becomes visible: the median onset is 5.0 at 20,000, 6.0 at 60,000, and 7.0 at 120,000 warmup gradients. Appendix B gives every current 60k row, the single-chain contrast, and the matched ablations.

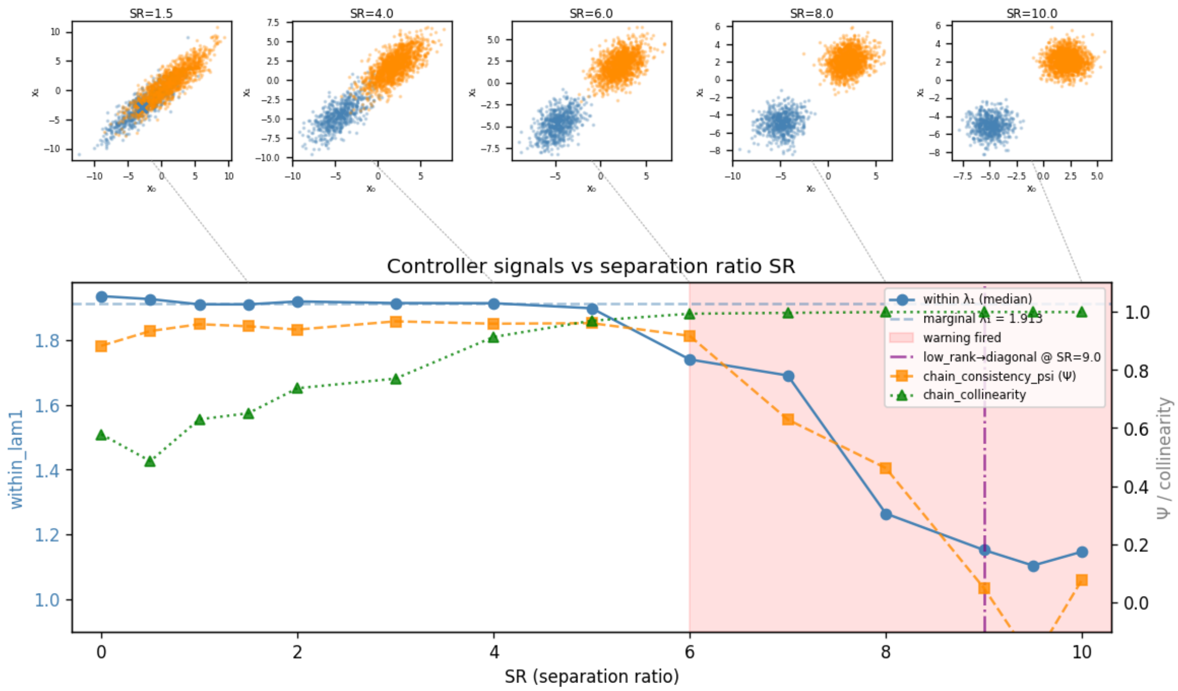

Figure 4: Current-corpus controlled GMM boundary, generated from the current $k = 2$ primary corpus. Top: target draws at selected separation ratios. Bottom: seed-median current diagnostics. The marginal spike remains 1.913; the shading begins at the first grid point with a persistent disagreement signal in any seed, and the dash-dot line marks the first all-diagonal grid point. These are current-window signals: a population handoff can clear, and `global_exploration` remains `not_established`.

6 Finite-horizon local transcripts

Structural evidence from visited states cannot by itself certify unseen regions. The appropriate obstruction is finite-horizon and depends on the joint event that every query remains local.

Theorem 6 (Local-transcript indistinguishability)**.** *Let q_0, q_1 be smooth, positive, integrable unnormalized targets. Target access is only through a declared oracle, and the two targets give identical oracle responses on an open region A. Run the same randomized algorithm through a fixed horizon T from the same initial law under both targets, couple all random numbers, and assume it makes finitely many oracle calls almost surely through T. Let E_T be the event that every oracle query made by the complete, possibly interacting algorithm through T lies in A. Induction over oracle queries then gives identical coupled transcripts on E_T and a common submeasure of mass*

$$p_T = P_0(E_T) = P_1(E_T).$$

If the normalized targets π_0, π_1 have metric functionals satisfying $d\{T(\pi_0), T(\pi_1)\} > 2\epsilon$, then every common transcript-measurable estimator $\widehat{T}$ satisfies

$$\Pr_0\{d(\widehat{T}, T(\pi_0)) > \epsilon\} + \Pr_1\{d(\widehat{T}, T(\pi_1)) > \epsilon\} \geq p_T. \tag{8}$$

Consequently the two-point minimax error is at least $p_T/2$.

For replica events $E_{T,m}$, the chain rule is

$$p_T = \prod_{m=1}^{M} P(E_{T,m} \mid E_{T,1} \cap \cdots \cap E_{T,m-1}).$$

Independent replicas give $p_T = \prod_m q_{T,m}$. If every displayed conditional probability is bounded below by q_T, then $p_T \geq q_T^M$. Identical independent replicas are sufficient for $p_T = q_T^M$ when each has confinement probability q_T, but they are not necessary: equality in the displayed lower bound occurs whenever every conditional factor equals q_T, including some dependent constructions. Shared adaptation keeps the joint p_T form. The lower bound applies when $M = 1$ and makes multiple chains neither mathematically necessary nor sufficient for discrimination. Genuinely overdispersed multiple chains remain a principal practical safeguard against metastability and unvisited modes. Longer runs, more starts, or a more diverse collection of starts can reduce the confinement event, but need not do so at fixed compute. No fixed sample count is privileged: its meaning is target-, kernel-, start-, and controller-specific unless p_T is bounded for a declared problem class. This is a finite-horizon Le Cam two-point coupling argument [Yu, 1997].

7 Routing and efficiency evidence

The scalar-gate implementation selected low rank in all 12/12 ill-conditioned and German-credit cells. Across automatic, Fisher, and diagonal arms, all 36/36 post-sampling population-quality checks passed: rank-normalized split-$\widehat{R}$ was finite with maximum at most 1.01, and sampling divergences were zero. This is a reporting rule for the observed population, not routing confidence or a global-exploration result. The corpus uses ArviZ rank-normalized split-$\widehat{R}$ [Vehtari et al., 2021]; 4,996 warmup divergences are retained rather than screened from the record.

Table 1 compares automatic warmup with the predeclared Fisher low-rank primary and, separately, the diagonal control. Automatic warmup beats the Fisher primary in every observed cell under all three ESS-per-gradient estimands. Every ratio charges gradients actually consumed. The pooled and marginal fixed-suite comparisons use $M = 8$ chains, float64 arithmetic, explicit integer seeds, and a 312-transition equal-split population comparator under the preregistered nominal allocation. Pooled accounting charges all warmup and sampling gradients; marginal accounting charges one chain's sampling gradients and $1/M$ of joint warmup, and its geometric means aggregate paired seed-by-chain ratios. The one-output column is descriptive: its comparator is the historical single-chain 2,500-warmup policy and is not budget-identical.

Table 1: Geometric-mean automatic/comparator ESS-per-gradient ratios under the column-specific accounting defined in the text. Fisher low rank is the predeclared primary; diagonal is a control and was not selected after observing the results.

Model	comparator	pooled population	marginal amortized	one output
ill-conditioned	Fisher low rank	2.451	2.324	1.409
ill-conditioned	diagonal control	22.572	21.124	14.196
German credit	Fisher low rank	1.951	1.882	1.131
German credit	diagonal control	6.264	5.887	6.009

When M chains share one step size, the M acceptance statistics observed at a single step size are replicated measurements for one controller time, not M successive controller times. The shared-step comparison contrasts the current mean-pooled bundle with revision `2f62921848a93e7dc544ba9de8e29ef177e373b6`, which applied those statistics as successive dual-averaging updates. The frozen historical/current ratios mean that the current bundle used 19.38–19.89 times fewer warmup gradients in the ill-conditioned cells and 31.82–35.60 times fewer for German credit. Mean pooling, the warmup tree-depth cap, and the controller semantics changed together, so these ratios belong to the bundle and do not identify the effect of pooling alone.

Off NUTS, low-rank selection occurred in 12/12 fixed-length and multinomial HMC cells with zero post-warmup divergences. This is route support for those cells, not uniform utility or a population-quality result. Multinomial HMC lost to its comparator in 2/6 cells. The fixed-length ratios are dominated in some cells by near-zero manual denominators and are not summarized as a general gain.

8 Open-source implementation

BlackJAX provides the composable inference framework used here [Cabezas et al., 2024]. The evaluated controller is the `blackjax/adaptation/meta` module at revision `29d2468857be4de1644ca4470c2a4aa7f8137656`.[1] `staged_adaptation` is the host interface that supplies the step-size controller, static schedule, and metric core.

```
warmup = staged_adaptation(
    blackjax.nuts, logdensity_fn,
    metric="auto",
    max_grad_budget=50_000,
    n_chains=8,
)
(last_state, parameters), info = warmup.run(rng_key, initial_positions)
```

The public interface exposes one compute budget and the chain population; Appendix D records the full signature.

[1] https://github.com/blackjax-devs/blackjax/tree/29d2468857be4de1644ca4470c2a4aa7f8137656/blackjax/adaptation/meta

9 Related work

Controlled-Markov stochastic approximation and adaptive-MCMC theory supply ingredients for stability, containment, transient control, and diminishing adaptation [Andrieu et al., 2005, Fort et al., 2016, Roberts and Rosenthal, 2007]. They do not by themselves establish the five-term finite-window update budget in (4). Markov CLTs and multivariate output analysis make LRV uncertainty functional-specific [Jones, 2004, Vats et al., 2018]. Matrix concentration under dependence is available under stronger boundedness and dependence hypotheses [Neeman et al., 2024].

Window adaptation [Carpenter et al., 2017], Fisher-divergence preconditioning [Seyboldt et al., 2026], and entropy-based adaptive HMC [Hirt et al., 2021] provide direct metric or mass-matrix adaptation. Hird and Livingstone [2025] characterize when constant linear preconditioning can and cannot improve conditioning. Pathfinder instead builds local Gaussian approximations along an L-BFGS path for variational inference and initialization [Zhang et al., 2022]. ChEES tunes HMC trajectory length [Hoffman et al., 2021], while MEADS adapts an ensemble of generalized HMC chains [Hoffman and Sountsov, 2022].

For regional exploration, sequential Monte Carlo supplies population-level reweighting, resampling, and mutation over a sequence of bridging distributions [Del Moral et al., 2006]; this does not make resampling alone a guarantee of mode crossing. Tempered transitions are a direct regional/tempering precedent [Neal, 1996]. The affine-invariant ensemble sampler is included as an example of robustness to affine scaling, not as evidence of separated-mode exploration [Goodman and Weare, 2010].

Query-complexity lower bounds ask how many oracle calls are required to produce a total-variation-accurate sample over declared smoothness classes [Chewi et al., 2022, He and Zhang, 2025]. The stochastic-gradient, strongly-log-concave case gives a related oracle lower bound [Chatterji et al., 2022]. Those results differ from Theorem 6, which lower-bounds estimation of a target functional from a finite local transcript through its confinement mass p_T.

The corpus reports rank-normalized split-$\widehat{R}$ in the sense of Vehtari et al. [2021]. Nested-$\widehat{R}$ addresses a distinct design: superchains share an initializer within groups and are conditionally independent [Margossian et al., 2025]. It does not validate an arbitrary post-hoc grouped rank-normalized statistic, and no such extension is used here.

The iid calibration uses spiked-PCA theory [Baik et al., 2005, Baik and Silverstein, 2006, Paul, 2007], while Onatski et al. [2013] cautions against turning the PCA transition into an unrestricted testing impossibility. Weyl, Davis–Kahan variants, and matrix- function perturbation results support only the conditional gate consequences stated in Theorem 5 [Bhatia, 1997, Yu et al., 2015, Higham, 2008].

10 Discussion

The route-attractor result makes metric convergence conditional on a route margin and an update-error budget. The scalar-gate implementation does not supply the confidence-aware design's confidence objects. Its verdict instead reports the metric's route and conditioning scope, observed ensemble disagreement, and advisory handoff. The state tree in Section 3 confines every such reading to the observed transcript.

The GMM boundary is compatible with the exact within/between decomposition but does not identify a unique occupation mechanism. Its $k = 3$ ablation exercises the deployed low-rank structure and finds no general efficiency gain. The finite-transcript result leaves both run length and starting diversity as ways to reduce the finite-horizon confinement probability p_T, subject to the actual interacting algorithm; it does not make multiple chains necessary.

Refusal remains an explicit output: funnels and scale coupling route to reparameterization, whereas genuinely regional mixtures route to the companion population/regional ensemble. Any selected metric remains available to the companion local-trajectory sampler. Fixed-L kernels use their declared integration length for gradient-budget accounting. Optional within-window termination could save work after evidence stabilizes, but it is not a prerequisite for the dimension/rank-capacity-indexed decision schedule evaluated here.

11 Conclusion

The Universal Warmup Path begins diagonal, gathers route-specific evidence, and then acts, waits, or refuses. Route-indexed attractors explain what happens after a supported decision; operator uncertainty and local-transcript limits govern when that decision is available. The empirical corpus shows that the automatic route can outperform its predeclared Fisher primary while still recording substantial warmup pathology, mixed schedule effects, and advisory rather than global mixture evidence. Refusal is the mechanism that keeps those local successes from becoming a global claim.

Generative-AI use disclosure

Anthropic Claude Opus 4.8 and Claude Fable 5, and OpenAI Codex and GPT-5.6, supported literature discovery and checking, proof critique and derivation, code generation and review, experiment orchestration and validation, and manuscript drafting and editing. Junpeng Lao directed the work, reviewed and revised model output, checked model-supported claims against cited primary sources, source code, and frozen experiment outputs, and accepts responsibility for the manuscript. No AI system is an author or a source.

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A Proofs and technical details

Proof of Lemma 1. On the event $\eta_k \in C_k$, the inclusion $C_k \subset H_b$ implies $\eta_k \in H_b$. Iterated expectation and a union bound over the fixed potential-window set $\mathcal{K}$ give $\Pr(\exists k \in \mathcal{K} : \eta_k \notin C_k) \leq \sum_{k\in\mathcal{K}} \delta_k$. This bound is taken before observing which windows trigger an opportunity. Every realized promotion is therefore supported on the complementary event. □

Proof of Proposition 3. If $T_a = F_a \circ \mathrm{Cov}$, equal covariance plainly implies equal route target. Conversely, if T_a is constant on every covariance fibre, define $F_a(\Sigma)$ to be $T_a(\pi)$ for any $\pi \in \mathcal{P}$ with $\mathrm{Cov}_\pi(X) = \Sigma$. Fibre constancy makes this definition independent of the representative. □

Proof of Proposition 4. Integration by parts under the Stein boundary conditions gives $E\{(X-\mu)s(X)^\top\} = -I$. Hence the covariance matrix of $(X-\mu, s(X))$ is

$$\begin{pmatrix} \Sigma_\pi & -I \\ -I & J_\pi \end{pmatrix} \succeq 0.$$

Its Schur complement gives $J_\pi \succeq \Sigma_\pi^{-1}$. The definition of the SPD geometric mean then yields

$$\Sigma_\pi^{-1/2}(\Sigma_\pi \# J_\pi^{-1})\Sigma_\pi^{-1/2} = (\Sigma_\pi^{1/2} J_\pi \Sigma_\pi^{1/2})^{-1/2} \preceq I.$$

Equality in the Schur complement makes $s(X) = -\Sigma_\pi^{-1}(X-\mu)$ almost surely. Integrating this affine score identifies the Gaussian density; the Gaussian attains equality. For the variance-one Student-t_ν example, direct integration of the squared location score gives $J_\pi = \nu(\nu+1)/\{(\nu-2)(\nu+3)\}$. □

Proof of Theorem 2. Write $d_k = \theta_k - \theta_a^\star$. Dissipativity and the growth bound give

$$\|d_k + \alpha h_a(\theta_k)\|_F^2 \leq (1 - 2\alpha\mu + \alpha^2 L_h^2)\|d_k\|_F^2.$$

For any $z \in \overline{B}_{\mathcal{S}}(\widehat{s}_k, r_k^s)$, the route inequalities give

$$\|z - s(\theta_a^\star)\|_{\mathcal{S}} \leq 2r_k^s + L_s\|\theta_k - \theta_a^\star\|_F < \kappa.$$

Because $s(\theta_a^\star)$ is interior to $\mathcal{S}_a$, this proves $\overline{B}_{\mathcal{S}}(\widehat{s}_k, r_k^s) \subset \mathcal{S}_a$. The candidate update is therefore selected without conditioning its error event on the route. Equation (2) then gives $\|d_{k+1}\|_F \leq q\|d_k\|_F + e_k$. The basin condition closes the induction, iteration yields (3), and a conditional union bound gives the stated probability. □

Proof of Theorem 5. The eigenvalue intervals are Weyl's inequality. The two strict inequalities around τ certify exactly k eigenvalues above the cutoff. Applying the Davis–Kahan sin-Θ theorem on an event with gap exceeding $2r$ gives the displayed conservative bound. Finally, the spectral floor keeps both matrices SPD along the perturbation path; the integral representation of the matrix logarithm bounds its derivative by the reciprocal spectral floor, giving $\log\{m/(m-r)\} \leq r/(m-r)$. None of these steps specifies the map from the residual moments to a route metric, hence the separate T_a condition. □

Proof of Theorem 6. The initial states and randomized choices are coupled. If the first $j-1$ queries lie in A, their oracle responses agree under q_0 and q_1, so the algorithm states and the jth query agree. Induction over all oracle calls therefore makes the complete transcripts identical on E_T and proves the common submeasure of mass p_T. On that submeasure the estimator output is the same. The two ϵ-balls around $T(\pi_0)$ and $T(\pi_1)$ are disjoint, so that output must be wrong for at least one target. Integrating over the common mass yields (8); taking the larger error gives the minimax bound. □

B Controlled GMM boundary study

The target construction pins the marginal diagonal-whitened top eigenvalue at 1.913. Here $\mathrm{SR} = \delta/s(\delta)$ is the separation ratio defined in §5. The primary 60k sweep is:

Table 2: Current 60k $k = 2$ sweep across three seeds. "Persistent" counts `persistent_disagreement_signal`; "handoff" counts the advisory `population` endpoint. Every row reports global exploration as `not_established`.

SR	low-rank/diagonal	persistent	handoff	within λ_1 range
0	3/0	0/3	0/3	1.910–1.979
0.5	3/0	0/3	0/3	1.916–1.932
1	3/0	0/3	0/3	1.907–1.952
1.5	3/0	0/3	0/3	1.907–1.962
2	3/0	0/3	0/3	1.895–1.937
3	3/0	0/3	0/3	1.882–1.929
4	3/0	0/3	0/3	1.913–1.920
5	3/0	0/3	0/3	1.842–1.907
6	3/0	2/3	2/3	1.737–1.847
7	3/0	3/3	2/3	1.371–1.707
8	2/1	2/3	2/3	1.205–1.266
9	0/3	2/3	2/3	1.093–1.297
9.5	0/3	2/3	2/3	1.102–1.258
10	1/2	2/3	2/3	1.081–1.219

Single-chain contrast. At the evaluated budget and seed, one chain selects low rank at $\mathrm{SR} = 1.5$, diagonal at $\mathrm{SR} = 5$, and diagonal at $\mathrm{SR} = 10$. The last row has mode-weight estimate 1.0. Ensemble evidence is unassessed for these single-chain rows, and global exploration is not established. These are scalar-gate- and budget-specific outcomes, not claims about every single-chain method or the necessity of multiple chains.

Matched diagonal ablation. The comparison holds warmup, step size, initialization, and $M = 8$ fixed. Its minimum-coordinate bulk-ESS-per-gradient low-rank/diagonal ratios are 0.869230, 0.997721, 1.170288, and 0.997752 at $\mathrm{SR} = 1.5, 4, 5, 9$, respectively. These four cells are descriptive: identification of anisotropy does not by itself establish a general efficiency gain.

Three-axis primary sweep. For $k = 3$, all three seeds report persistent disagreement from the first evaluated $\mathrm{SR} = 6$ grid point onward. The endpoint population handoff is nevertheless 2/3 at each of those separation ratios and can clear within a run. The persistent evidence field and current-window handoff therefore answer different questions.

Three-axis matched ablation (null). A clean same-M paired ablation used $k = 3$ correlated axes[2] at $\mathrm{SR} \in \{3.0, 3.5, 4.0, 4.5\}$, seeds 1001–1016, and $M = 8$. Both arms shared the 60k warmup, terminal state, and step size; each then produced 4,000 draws per chain. For equal-SR, within-seed pairs, the low-rank/matched-diagonal projection bulk $\mathrm{ESS}(v^\top x)$ per gradient geometric-mean ratio was 1.037; the seed-cluster t 95% interval was $[0.998, 1.077]$. Because the interval includes 1, the stated measurable, non-noise go/no-go criterion fails: this is a null result and supports no efficiency claim. Per-SR geometric means were 1.002, 0.999, 0.987, and 1.169 in ascending SR order; the $\mathrm{SR} = 4.5$ rise is local and does not generalize. All 64/64 runs completed, with 16/16 valid low-rank profiles at each SR, zero divergences, and maximum projection split-$\widehat{R}$ 1.006. Of the 64 paired ratios, 39 are above 1 and 25 below.

[2]For v supported equally on k axes, diagonal whitening gives $\lambda_1(k) = (1 + a)/(1 + a/k)$. For $k = 2, \dots, 5$, the marginal spikes are $\{1.913, 2.750, 3.520, 4.231\}$ and the off-diagonal counts are $\{1, 3, 6, 10\}$, respectively. For $k = 2$, the scalar-gate implementation's asymptotic covariance-spike comparison is $1.913 < 2$, where 2 is its escalation threshold; this is neither a universal impossibility nor a finite-budget guarantee. At $k = 3$, $2.750 > 2$, so that particular threshold contrast is absent.

C Schedule-comparison scope

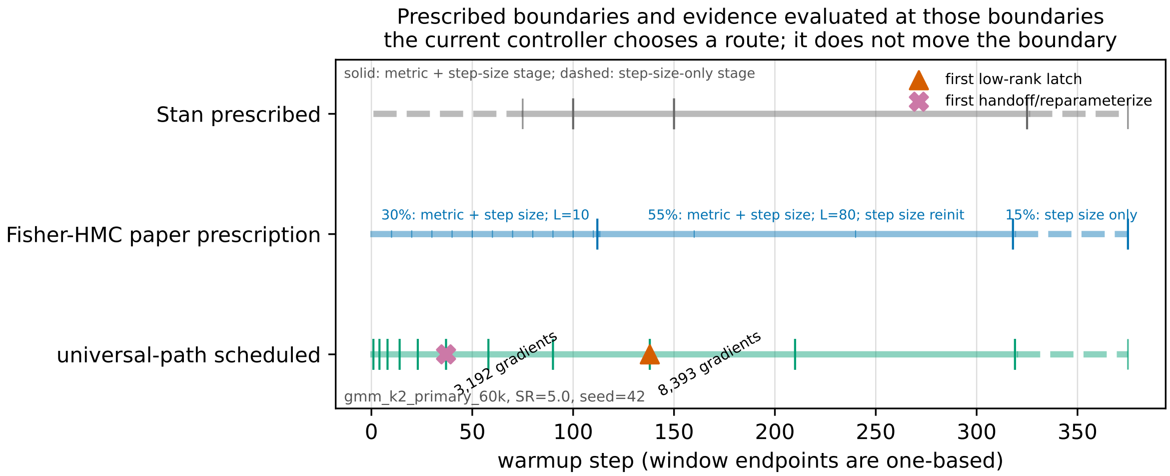

Figure 5: Schedule evidence and one executed controller trace. The Stan and Fisher-HMC lanes are unexecuted reference prescriptions; Fisher-HMC denotes the specific 30/55/15 prescription in Seyboldt et al. [2026]. Universal-path route decisions occur only at prescribed, unmoved boundaries: the scalar-gate implementation does not fit change-point timing. In the displayed trace the first population-handoff marker is transient and clears; terminal global exploration remains `not_established`.

Full factorial. The validated schedule-by-buffer experiment reveals a severe failure on radon for the *bundle* combining proportional-growing windows with an accumulating/per-draw-recompute buffer. Because accumulation and recomputation move together in this arm, the result does not identify accumulation alone as the cause. The remaining cells show that schedule, buffer, and controller semantics must be compared as configurations rather than as a pure window-shape effect. The Universal-path boundaries in Figure 5 are prescribed before execution and are not moved by detected events.

Restart ablation. Continuous dual averaging has mixed efficiency: it wins 3/6 cells, with continuous/reseed ratios from 0.9464 to 1.1385 and geometric means 1.014 for the ill-conditioned target and 1.024 for German credit. It reduces the recorded oscillation and warmup-divergence burden, but this limited ablation establishes neither directional efficiency nor equivalence between Stan and `nutpie`.

D Full warmup signature

```
staged_adaptation(
  algorithm, logdensity_fn, metric="welford_diag" | MetricRecipe | MetricCore,
  *, max_grad_budget=None, n_chains=1, imm_shrinkage_to_previous=0.0,
  initial_inverse_mass_matrix=None, initial_step_size=1.0,
  target_acceptance_rate=0.80, adaptation_info_fn=..., integrator=velocity_verlet,
  schedule_fn=None, initial_metric_state=None, **extra_parameters)
```

Step-size dual averaging and the stage schedule live in the host; mass-matrix estimation is delegated to `metric`.